

FITCHBURG, MA, USA

WMO: 725107

Lat: 42.552N

Lon: 71.756W

Elev: 348

StdP: 14.51

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04780

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.3	7.5	-11.9	2.9	6.4	-7.7	3.7	11.0	25.4	24.8	22.9	23.5	6.9	310	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	90.9	72.6	88.2	71.2	85.4	69.5	75.5	85.4	74.1	83.2	72.6	81.2	9.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.7	122.9	79.7	71.6	118.2	78.8	69.9	111.5	77.8	39.2	85.5	37.7	82.7	36.4	81.3	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.5	16.4	DB	-4.2	96.0	4.6	3.1	-7.6	98.2	-10.3	100.0	-12.9	101.8	-16.3	104.0	(4)
(5)			WB	-5.0	78.1	4.6	1.5	-8.3	79.2	-11.0	80.1	-13.6	80.9	-16.9	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	25.8	28.1	36.2	48.3	58.7	67.6	73.3	71.5	64.4	51.9	41.5	31.5	(6)	
	DBStd	18.45	10.52	9.54	9.35	8.68	8.06	7.14	5.71	5.65	7.14	8.13	8.48	9.03	(7)	
	HDD50	2880	751	616	440	133	12	0	0	0	0	75	278	575	(8)	
	HDD65	6204	1215	1034	894	507	232	53	6	10	98	413	705	1037	(9)	
	CDD50	2886	1	2	12	82	282	527	722	668	431	133	23	3	(10)	
	CDD65	734	0	0	1	6	37	130	262	213	79	6	0	0	(11)	
	CDH74	6918	0	1	8	109	480	1251	2475	1852	682	60	0	0	(12)	
	CDH80	2313	0	0	1	35	159	416	917	583	196	6	0	0	(13)	
(14) Wind	WSAvg	6.2	6.9	7.2	7.5	7.4	6.4	5.8	5.5	5.1	4.9	5.6	6.1	6.4	(14)	
(15) Precipitation	PrecAvg	45.2	3.4	3.3	3.9	3.5	3.5	3.8	3.8	3.8	3.8	4.0	3.7	4.1	(15)	
	PrecMax	59.8	6.6	10.7	10.4	8.0	7.0	8.8	8.7	8.9	8.6	13.4	6.8	7.5	(16)	
	PrecMin	32.8	1.2	1.4	0.6	0.6	0.8	0.6	1.5	0.5	0.8	0.1	1.1	1.3	(17)	
	PrecStd	6.8	1.3	1.7	1.9	2.0	1.5	2.1	1.8	2.0	2.2	3.0	1.3	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	62.0	72.1	85.0	90.3	92.7	94.6	92.9	90.5	79.8	69.6	63.3	(19)	
		MCWB	55.3	52.9	56.4	62.3	67.1	73.2	74.9	74.8	72.8	64.7	59.5	56.9	(20)	
	2%	DB	51.6	53.4	62.9	75.5	84.4	88.5	91.0	89.4	85.5	74.9	64.6	54.6	(21)	
		MCWB	45.9	46.2	51.7	56.7	65.0	71.1	73.2	72.7	70.1	64.3	56.7	49.6	(22)	
	5%	DB	45.0	47.4	55.9	69.9	79.4	84.5	88.2	86.0	81.3	70.6	60.7	49.7	(23)	
		MCWB	40.1	40.4	45.8	54.7	62.2	68.3	71.8	70.6	67.6	61.9	52.7	44.2	(24)	
	10%	DB	39.6	41.9	50.5	64.4	74.5	81.3	85.2	83.0	77.4	66.0	56.2	45.0	(25)	
		MCWB	34.8	35.9	42.5	51.8	60.3	66.8	70.2	69.3	66.3	57.7	49.6	40.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	56.3	58.9	65.3	70.7	75.7	78.3	77.2	75.2	70.0	63.3	57.7	(27)	
		MCDB	58.3	60.8	68.2	77.0	84.9	87.6	89.9	87.9	85.6	74.7	67.3	61.1	(28)	
	2%	WB	46.5	47.1	53.4	61.1	68.1	73.7	76.0	75.2	72.6	66.6	58.4	50.7	(29)	
		MCDB	49.8	52.4	61.5	72.8	79.6	83.7	86.1	84.1	81.1	72.1	62.8	54.1	(30)	
	5%	WB	40.3	40.5	47.7	57.3	65.6	71.8	74.2	73.8	70.8	62.6	54.4	45.4	(31)	
		MCDB	45.1	46.2	54.4	67.0	75.1	80.8	83.6	81.5	77.9	69.4	58.6	48.4	(32)	
	10%	WB	35.6	36.6	43.1	53.0	62.7	69.5	72.6	72.1	68.9	59.3	50.4	40.5	(33)	
		MCDB	38.9	41.1	49.7	62.3	71.2	77.7	81.5	79.7	75.2	65.0	55.9	44.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	18.0	18.3	20.9	21.0	19.6	19.6	19.7	20.3	19.6	18.0	15.6	(35)	
		MCDBR	20.9	24.8	26.9	30.2	28.5	23.6	22.2	21.9	23.6	24.9	24.1	21.6	(36)	
	5% WB	MCWBR	17.4	18.3	16.9	15.5	13.3	10.3	8.7	8.8	11.1	14.5	16.7	18.0	(37)	
		MCWBR	20.4	22.9	24.2	26.3	24.1	20.8	19.6	18.3	19.4	21.0	20.2	20.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.284	0.295	0.324	0.366	0.402	0.441	0.461	0.428	0.372	0.331	0.305	0.286	(40)	
		taud	2.456	2.436	2.448	2.379	2.315	2.233	2.185	2.262	2.397	2.530	2.556	2.545	(41)	
	Ebn at Noon		275	290	293	286	277	266	258	264	272	273	266	266	(42)	
		Edh at Noon	23	28	32	36	40	44	45	41	33	26	22	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		568	840	1159	1469	1617	1751	1853	1666	1348	876	609	463	(44)	
	RadStd		36	67	79	157	193	171	113	84	95	82	52	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+1.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+115	(46)	
(47)	Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+70	(47)	

Nomenclature: See separate page